Introduction to Python

A few definitions

- Algorithm - A sequence of well-defined instructions in a particular order, used to solve a problem or perform a task
- Programming language - A language that allows a computer to understand instructions
- Computer program - A sequence of well-defined instructions in a particular order, that tells a computer how to perform a task. In other words, an implementation of an algorithm in a programming language

Python is already on most computers

- Python comes pre-installed on most modern computers
- But not on the class set of laptops
- On your computer you can type “python3 --version" in a command line interface (CMD on Windows, Terminal on Mac) to see what version of python your computer has

But it’s complicated

- You have to edit your programs in a text editor, save as a .py file, then open with python
- If you want to edit a program you just ran, you have to go back to the text editor, then back to python
- We would prefer some sort of Integrated Development Environment in which you edit your program and run it from the same environment

Integrated Development Environments

- There are multiple different IDEs available — some are free, and some you pay for
- A very popular free IDE is Visual Studio Code from Microsoft
- But so that you guys don't need to download anything, and to solve the issue of things that we show in class might work on one computer and not another, we will use a browser based IDE for this class
- Github Codespaces (on github.com) is a free-ish browser-based IDE with a Visual Studio backend (60 hours per month free)
- You may use a personal email or Kinkaid email to signup for github. A confirmation code will be sent to the email that you choose. If using a Kinkaid email, you access the email from gmail.com and type in your login credentials.

Hello World

- Since the 1970s, there has been a tradition that the first program written when learning a new programming language is a program that displays the text “Hello World!”
- I am providing a template in GitHub with just such a program
- https://github.com/nhammen-kinkaid/python-lesson-1
- Open hello_world.py, and run it
- We can change the text in the print statement, or use a number rather than a string of text. Try things out.
- Note that a new line in python code is a new statement / instruction

Variables

- A variable is an abstract container that stores data
- The data currently stored in a variable is called its value
- Open variables.py and we will discuss it
- Dynamically typed vs statically typed:
  - Python is dynamically typed, which means that any type of data can be put in any variable
  - Some languages are statically typed, which means that each variable has specific types of data they can hold, and an error occurs if the wrong type of data is put in
- When a new value is given to a variable, this overwrites the old value.
- Variable names are case-sensitive

Getting input from a user

- We have already used print() to display output to a user
- We can use input() to get input from a user
- <var> = input(<str>) will display <str> and then wait for the user to give input. Once the user enters a new line, everything before the new line will be put into <var>
- A short example is contained in input.py
- What happens if we enter a number?